SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	1	**Date:**	11/10/2026

Code of Conduct

I Tejhash P ,192512495 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejhash.P

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Answer – Question 1: Two-Room Vacuum World

(a) PEAS description

Component	Description
Performance measure	+1 for every clean square at the end of each time step; −p for every Left or Right move; evaluated over T = 6 steps
Environment	Two squares A and B, each Clean or Dirty; agent starts in A with both squares Dirty; cleaned squares stay clean
Actuators	Suck, Right, Left, NoOp (wheels and suction)
Sensors	Location sensor and a dirt sensor for the current square, giving the percept [location, Dirty/Clean]

(b) Agent X (simple reflex), p = 1

Step	Percept	Action	Clean squares at end of step	Move penalty	Step score
1	[A, Dirty]	Suck	1 (A)	0	+1
2	[A, Clean]	Right	1 (A)	−1	0
3	[B, Dirty]	Suck	2 (A, B)	0	+2
4	[B, Clean]	Left	2	−1	+1
5	[A, Clean]	Right	2	−1	+1
6	[B, Clean]	Left	2	−1	+1
Total			10	−4	**6**

Total for X = 1 + 0 + 2 + 1 + 1 + 1 = **6**. X keeps shuttling between the squares after both are clean.

(c) Agent Y (reflex with internal state), p = 1

Step	Percept	Action	Clean squares at end of step	Move penalty	Step score
1	[A, Dirty]	Suck	1 (A)	0	+1
2	[A, Clean]	Right	1 (A)	−1	0
3	[B, Dirty]	Suck	2 (A, B)	0	+2
4	[B, Clean]	NoOp	2	0	+2
5	[B, Clean]	NoOp	2	0	+2
6	[B, Clean]	NoOp	2	0	+2
Total			10	−1	**9**

Total for Y = 1 + 0 + 2 + 2 + 2 + 2 = **9**. After step 3 Y remembers that both squares have been cleaned, so it stops moving.

(d) Scores as functions of p

- Sum of clean-square points is the same for both agents: 1 + 1 + 2 + 2 + 2 + 2 = 10.
- X makes 4 moves (steps 2, 4, 5, 6): $\mathbf{S_X = 10 - 4p}$. Y makes 1 move (step 2): $\mathbf{S_Y = 10 - p}$.
- $S_Y - S_X = 3p$, so Y is strictly better than X when $\mathbf{p > 0}$. At p = 0 they tie (10 each); for p < 0 (moving rewarded) X would be better. Check at p = 1: 6 and 9.

(e) Interpretation

Which agent is rational? For any p > 0, Y is rational: for each percept sequence it picks the action expected to maximise the performance measure, and once both squares are clean the best action is NoOp because every move costs p. X is not rational for p > 0, since it keeps paying move penalties for no gain. At p = 0 both are equally rational.

Role of the performance measure and percept history. Rationality is judged against the performance measure (here moves are penalised) and against everything the agent has perceived so far, not the current percept alone.

Why X cannot match Y. Both agents receive the same percepts up to step 4, including [B, Clean]. That percept alone cannot tell whether A has been cleaned. X's agent function uses only the current percept, so [B, Clean] always means "move". Y's internal state records that A and B are clean, so it chooses NoOp. Y acts on the percept history, X cannot.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/ Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Answer – Question 2: Task Environment Analysis and Agent Selection

(a) Classification with justification (score in brackets)

Property	E1: Autonomous taxi	E2: Part-picking robot	E3: Chess with clock
Observability	Partially observable: cannot see round corners or other drivers' intentions (1)	Partially observable: camera sees only a limited view; defects and occlusions are uncertain (1)	Fully observable: the whole board is visible (0)
Number of agents	Multi-agent: other vehicles and pedestrians (1)	Single agent: only the robot acts on the belt (0)	Multi-agent: an opponent (1)
Determinism	Stochastic: traffic and road events are unpredictable (1)	Stochastic: noisy vision, slipping grasps, random arrivals (1)	Deterministic: each move has a certain outcome (0)
Episodic / Sequential	Sequential: current driving decisions affect later situations (1)	Episodic: each part is judged independently of earlier ones (0)	Sequential: every move shapes the rest of the game (1)
Static / Dynamic	Dynamic: traffic moves while the taxi decides (1)	Dynamic: the belt keeps moving while the robot decides (1)	Semi-dynamic: board does not change but the clock runs down the score (0.5)
Discrete / Continuous	Continuous: position, speed, steering are continuous (1)	Continuous: image and arm positions are continuous (1)	Discrete: finite board states and moves (0)

(b) Difficulty scores and ranking

Environment	Observ.	Agents	Determ.	Episodic/Seq.	Static/Dyn.	Disc./Cont.	Total
E1 Taxi	1	1	1	1	1	1	**6**
E2 Part-picking robot	1	0	1	0	1	1	**4**
E3 Chess with clock	0	1	0	1	0.5	0	**2.5**

Ranking from hardest to easiest: **E1 (6) > E2 (4) > E3 (2.5)**.

(c) Agent architecture for the hardest and easiest environments

E1 (hardest): utility-based agent (with an internal model, and learning to improve it). The environment is partially observable, stochastic, dynamic and multi-agent, so the agent must keep an internal state of unseen road conditions and predict other drivers. Goals conflict (safe, fast, legal, comfortable) and outcomes are uncertain, so it needs expected utility to trade them off; a simple reflex agent would fail.

E3 (easiest): goal-based agent using search. The environment is fully observable, deterministic and discrete, so the agent can simulate the exact result of each move from the rules and search for moves that lead to the goal (checkmate), using an evaluation function and the clock for time management. It needs no model of hidden state or probabilities.

(d) Interpretation

The complexity of the agent program depends on the properties of the environment, not just its domain. Partial observability forces an internal state, stochastic outcomes force reasoning about probabilities and utilities, a dynamic setting forces fast decisions as the world changes, and a sequential setting forces planning ahead. Multi-agent settings require predicting others. A taxi and a chess player are both "agents in a task", but fully observable, deterministic, discrete chess allows a far simpler program than the open road.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Answer – Question 3: Turing Test Data Across the History of AI

(a) Correct-identification rate and fooling rate

System	Interrogations	Correctly identified	Correct-identification rate	Machine taken for human	Fooling rate
ELIZA	50	47	47/50 = 94%	50 − 47 = 3	3/50 = **6%**
Chatbot B	50	36	36/50 = 72%	50 − 36 = 14	14/50 = **28%**
System C	50	29	29/50 = 58%	50 − 29 = 21	21/50 = **42%**

Fooling rate = 100% − correct-identification rate.

(b) Turing's 70% criterion

Method: compare each system's correct-identification rate with 70%; a system passes if the rate is at most 70% (fooling rate at least 30%). Distance from chance is the rate minus 50 percentage points (pp).

System	Correct-id. rate	≤ 70%?	Gap to 70%	Distance from 50% chance
ELIZA	94%	No (fails)	24 pp above	+44 pp
Chatbot B	72%	No (fails, narrowly)	2 pp above	+22 pp
System C	58%	Yes (passes)	12 pp below	+8 pp

Conclusion: only System C satisfies the criterion. With only 50 trials the standard error is about 7 pp, so B's 2 pp miss is within noise, and C's 8 pp above chance is barely more than one standard error, meaning judges can hardly do better than guessing.

(c) Improvement in fooling rate

- ELIZA to System C: 42% − 6% = **36 percentage points**; ratio 42 / 6 = **7 times**.
- Chatbot B to System C: 42% − 28% = **14 percentage points**.

(d) Interpretation

The Turing Test measures only *acting humanly*: whether outward behaviour is indistinguishable from a human's. A high fooling rate therefore shows convincing imitation, not thinking or rational action. Using the four approaches to AI, passing sits in the "acting humanly" cell and says nothing about "thinking humanly" (cognitive models), "thinking rationally" (logic) or "acting rationally" (maximising expected performance). Searle's Chinese Room argument makes the same point: a system can manipulate symbols so well that it produces fluent replies without understanding them. Also, imitating humans includes copying human errors, which is not the same as acting rationally, so System C's 42% proves neither thought nor rationality.

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Answer – Question 4: Rational Route Choice under Uncertainty

U = 100 − T − 20·L, with L = 1 if T > 55 (late), otherwise L = 0.

(a) Utility of each route in each outcome

Route	Outcome	Time T (min)	Late? (L)	Utility U
R1 Highway	Normal	30	No (0)	100 − 30 − 0 = **70**
R1 Highway	Disrupted (accident)	70	**Late (1)**	100 − 70 − 20 = **10**
R2 City road	Normal	45	No (0)	100 − 45 − 0 = **55**
R2 City road	Disrupted (signal failure)	60	**Late (1)**	100 − 60 − 20 = **20**
R3 Ring road	Always	50	No (0)	100 − 50 − 0 = **50**

(b) Expected utilities and best route

- EU(R1) = 0.7 × 70 + 0.3 × 10 = 49 + 3 = **52**
- EU(R2) = 0.9 × 55 + 0.1 × 20 = 49.5 + 2 = **51.5**
- EU(R3) = 1.0 × 50 = **50**

The rational agent chooses **R1 (the highway)**, which has the highest expected utility (52).

(c) After the radio alert (P(disruption on R1) = 0.80)

EU(R1) = 0.2 × 70 + 0.8 × 10 = 14 + 8 = **22**. R2 (51.5) and R3 (50) are unchanged, so the agent now chooses **R2 (the city road)**.

(d) Outcome of the part (b) choice if disruption occurs

The agent took R1 and the accident happened: U = 100 − 70 − 20 = **10**, a late arrival (well below the expected 52). The agent was **not irrational**: it chose the action with the best expected utility given what it knew; a bad outcome from a sound decision does not make it irrational.

Rational vs omniscient: a rational agent maximises *expected* performance given its percept sequence and knowledge. An omniscient agent knows the actual outcome of each action in advance and would never pick a route that turns out badly. Omniscience is impossible in practice, and rationality does not require it.

(e) Interpretation

Rationality is defined relative to the performance measure (here time plus a heavy lateness penalty) and the percept sequence (the information the agent has received). The radio alert is a new percept that changes the agent's probability estimate for R1, even though the roads are physically the same. Expected utility of R1 is 70 − 60q for disruption probability q; it beats R2 only while q < 0.31 (51.5 = 70 − 60q gives q ≈ 0.308). At q = 0.8 the highway is no longer the best choice, so the rational decision changes with the information.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____